

GROUND WATER STATUS OF TAMILNADU

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ABSTRACT

Water is inevitable as it is an elixir and life sustaining liquid. Ground water forms a vital natural resources. Ground water is the water that seeps through rock and soil as it is stored in aquifers. Aquifers are typically made up of gravel, sand and sand stones. Being the largest liquid fresh water resources of the planet earth it supports irrigations to agriculture to lead a sustained life and secure food for the globe. India covers more than 30 per cent of the global irrigation and consumption. Based on the above background, an attempt has been made to study the ground water management and challenges in India and Tamilnadu.

Key words: Ground water, Aquifers, utilisable

INTRODUCTION

Water is inevitable as it is an elixir and life sustaining liquid. Ground water forms a vital natural resources. Ground water is the water that seeps through rock and soil as it is stored in aquifers. Aquifers are typically made up of gravel, sand and sand stones. Being the largest liquid fresh water resources of the planet earth it supports irrigations to agriculture to lead a sustained life and secure food for the globe. India covers more than 30 per cent of the global irrigation and consumption.

Sustainable use of ground water resources is very challenging as there is a high unplanned and non-scientific development of ground water. The alarming bell already sound that water level is depleting. Consequently developing ground water resources for sustainable uses and management is a great challenge.

Based on the above background, an attempt has been made to study the ground water management and challenges in India and Tamilnadu.

OBJECTIVES

The following objectives have been framed for the purpose of the study

1. To study the ground water resources in India and Tamilnadu
2. To identify the reasons for ground water depletion and give suitable suggestions

DATA AND METHODOLOGY

The present study is based on secondary data. Linear Trend analysis and Compound Growth rate has been used to analyse the secondary data.

ANALYSIS AND FINDINGS OF THE STUDY

GROUND WATER IN INDIA

Ground water is the sources and water is able unconfined and preched springs and aquifers controlling water bearing capacity of rocks. Ground water is the most extracted raw material with withdrawal rates currently estimated as 982 km³/year throughout the globe. About 70 per cent of groundwater withdrawn worldwide is used for agriculture. The United Nations Educational, Scientific and cultural organisations (UNESCO) report states that Indian is the largest that is second largest extractor of ground water in the world. It is predicted that by 2030, the 21 states in our nation are expected to run out of ground water. Trend analysis has been used to analyse the district wise classification of ground water level from 2014 to 2020.

TABLE 1
Ground Water Status in Different Countries

Country	Population (x 1000)	Area (km ²)	Ground water use (km ³ /yr)	Ground water use per person (m ³ /yr)	Ground water use per area (m ³ /km ² /yr)	Usage		
						Irrigation	Domestic	Industry
India	1,224,614	3,287,260	251.00	205	76,355	89%	9%	2%
China	1,341,355	9,596,960	111.95	83	11,665	54%	20%	2%
USA	310,384	9,826,680	111.70	360	11,367	71%	23%	6%
Pakistan	173,593	796,095	30.21	203	209,795	86%	13%	1%
Iran	73,974	1,648,200	29.45	260	14,992	72%	22%	6%
Bangladesh	148,692	143,998	24.24	883	11,276	92%	5%	3%
Mexico	113,423	1,964,380	29.45	260	14,992	72%	22%	6%
Saudi Arabia	27,448	2,149,690	24.24	883	11,276	92%	5%	3%
Indonesia	239,871	1,904,570	14.93	62	7,839	2%	93%	5%
Turkey	72,752	783,562	13.22	182	16,872	60%	32%	8%
Russia	142,985	17,098,200	11.62	81	680	3%	79%	18%
Syria	20,411	185,180	11.29	553	60,968	90%	5%	5%
Japan	126,536	377,915	10.94	86	28,948	23%	29%	48%
Thailand	69,122	513,120	10.74	155	20,931	14%	60%	26%
Italy	60,551	301,340	10.44	172	34,513	67%	23%	10%
Korea (South)	51,360	99,720	3.91	76	39,210	49%	46%	5%

Source: www.mowr.gov.in

It is clear from Table 1 that among the different countries, the usage of ground water usage is high in India. India, 89 per cent water is used for irrigation, 9 per cent is used for domestic and 2 per cent is used for industry.

TABLE 2
Ground Water Resources Availability and Status of its Utilisation in India

S.No.	Regions	Annual Replensishable Ground Water Resource (bcm)	Natural discharge during non- monsoon season	Net annual ground water available (bcm)	Annual ground water draft (bcm)	Stage of ground water development (%)	Categorisation units (Block)	
							Total assessment units	Block
1	2	3	4	5	6	7	8	9
2	Northern	5.4	0.48	4.92	1.84	37	30	

	Himalayan States							
3	North Eastern Hilly States	33.99	3.02	30.98	5.63	18	118	
4	Eastern Plain States	111.63	9.03	102.5	43.97	43	1895	
5	North Western Plain States	80.78	6.92	73.85	72.17	98	277	2
6	Western arid Region	27.38	1.97	25.4	24.48	96	462	1
7	Central Plateau States	90.723	5.19	85.53	36.11	42	985	
8	Southern Peninsular States	82.78	7.14	75.65	46.4	61	1946	4
9	Islands	0.34	0.01	0.32	0.01	4	10	
	Country total	433.02	33.77	399.26	230.63	58	839	

Source: www.cgwb.gov.in

It is clear from Table 2 that annual replenishable ground water resource is 433.02 bcm, natural discharge during monsoon season is 33.77 bcm and stage of ground water development is 58 per cent. Eastern plain states have majority rain fall when compared to other areas.

GROUND WATER STATUS IN TAMILNADU

Ground water is most important sources of fresh water in many regions of the world, especially in Tamilnadu its plays a predominant role in irrigation. The utilisable ground water discharge is 22,423 mcm. The current level of utilisation is of net ground water usage is 13.558 mcm that is about 60 per cent. Nearly 7 per cent of the total area of the state is occupied by a variety of land and fissured crystalline rocks like charnockite, peisses and granites.

TABLE 3

District Wise Classification of Ground Water Level from 2014 to 2020

Districts	2014	2015	2016	2017	2018	2019	2020
Thiruvallur	4.96	2.11	4.82	3.65	5.29	5.40	3.96
Kanchipuram	3.84	0.82	3.87	1.45	3.84	3.88	2.04
Thiruvannamalai	6.93	2.62	7.59	2.49	6.29	6.90	4.05
Vellore	8.29	3.35	8.5	5.54	8.14	8.73	7.27
Dharmapuri	8.31	4.02	8.76	8.64	12.13	12.41	10.75
Krishnagiri	6.01	5.81	9.79	7.71	8.47	8.47	8.06
Cuddalore	6.2	2.94	6.39	5.51	5.48	5.67	3.72
Villupuram	5.66	1.73	7.62	4.81	6.65	6.71	3.93
Thanjavur	4.58	2.78	4.57	2.75	2.32	2.54	2.44
Thiruvarur	2.68	0.72	3.2	1.79	2.52	2.90	2.12
Nagapattinam	1.52	1.02	3.74	4.33	2.89	2.82	1.55
Trichy	8.15	6.98	9.8	8.82	10.17	9.80	7.72
Karur	4.51	4.35	7.12	4.88	5.70	5.89	5.52
Perambalur	6.24	5.2	8.83	6.74	11.06	11.02	6.92
Pudukkottai	6.74	5.57	8.12	7.32	6.53	6.50	4.47
Ariyalur	3.88	2.19	3.69	2.86	5.85	5.77	2.40
Salem	10.03	7.43	12.25	7.88	11.62	11.61	6.15
Namakkal	11	7.8	12.69	9.46	10.56	11.48	9.57

Erode	6.23	5.56	12.02	10.44	7.27	7.67	6.14
Coimbatore	11.67	11.61	16.08	15.59	12.34	12.03	10.86
Thiruppur	10.07	8.92	12.62	10.79	9.82	9.42	9.21
Nilgiris	2.34	2.26	2.11	1.82	2.17	2.19	2.01
Dindigul	8.69	7.26	10.92	9.46	10.80	9.47	9.84
Madurai	6.21	5.63	8.42	7.76	6.26	6.84	5.82
Ramanathapuram	4.04	3.91	5.29	5.32	4.60	4.94	3.79
Sivagangai	5.15	5.49	7.54	8.34	8.38	8.34	4.99
Theni	10.98	8.14	12.1	9.06	8.70	8.18	8.50
Thoothukudi	4.25	3.51	6.15	5.66	5.09	5.31	2.35
Tirunelveli	4.02	1.75	7.49	5.94	4.57	4.44	2.76
Virudhunagar	8.2	6.17	9.27	8.17	8.21	8.64	7.96
Kanyakumari	4.87	2.77	5.55	3.42	4.84	5.24	5.10
a	5.8104	2.8086	6.6631	4.4714	6.6943	6.9074	4.9325
b	0.0325	0.1076	0.0814	0.1205	0.0223	0.0143	0.0384
R ²	0.0124	0.1353	0.0494	0.1173	0.0046	0.002	0.0153

Source: cgwb.gov.in/state_gw_departments.html

In the year 2015, the ground water level in all the districts is high as its R² value is high (0.1353) when compared to other years. In the year 2019, the ground water level is fall as its R² value is very low when compared to other years.

CAUSES OF DEPLETION OF GROUND WATER

The diversity in hydro-geologic set up and climatic conditions of the country has led heterogeneous recharge of natural available ground water. Inefficient water use practices, left irrigation system with poor maintenance have led to its depletion. The following reasons are identified as causes of depletion:

1. High Demand for Water

The demand of water has been increased for domestic, industrial and agricultural heads, but the surface water resources lead to the over exploitation of ground water.

2. Lack of Storage Facilities

Especially in hand-soak terrain along with lack of rain leads to depletion of water and also lack in storage facilities.

3. Green Revolutions

After green revolution during 1960, the walls for intensive crops and over dosage of fertilisers lead to over extraction of water.

4. Frequent Extraction

Continuous pumping of water from the ground without waiting for its replenishment leads to quick depletion of water.

5. Water Contamination

Water contamination as in the case of pollution by land fills, septic tanks, leaky underground gas tanks and over usage of fertilisers and pesticides lead to damage/depletion of ground water.

6. Climatic Changes

Water, the life sustaining liquid in depleting due to lack of rain and changes in hydro-geologic set up.

With increasing demand for water increases and lack of rain water led to depletion of ground water.

GOVERNMENT PROGRAMMES TO CONSERVE WATER

Water conservation is a n integral part of Indian identify and cultural history. In India, water conservation in the term of rain water harvesting is ancient tradition dates back to 4500 BC. Several efforts have been taken by the Government of India to conserve ground water, they are as follows:

1. Creation of new Ministry of Jal Shakti for dealing with all water ralting to wall.
2. National water policy has been formulated by Department of Water Resources.
3. Central Ground Water Authority (CGWA) has created a mandatory of rain water
4. Department of water resources, RD & GR has instituted National Water Awards to incentive good practices in water conservation and ground water recharge.
5. Mass awareness programmes are conducted from time to time each year under Information, Education and Communication scheme

SUGGESTIONS TO CONSERVE GROUND WATER

Protecting and conserving water is everyone's duty. The following measures can be taken to avoid depletion and conservation of ground water.

1. Construction, maintenance of public tube wells
2. Problems and issues such as water logging, agriculture toxins and industrial effluents to be treated properly.
3. Use of government schemes like DRIP programme, KishuSinchaiYojana and micro irrigation techniques can be used to avoid exploitation of water.
4. Bottom-up approach to empower local community and to create awareness.
5. Artificial recharge of tube wells, water reuse, afforestation, scientific method of agriculture is to be done.

CONCLUSION

Ground water is an important source of natural resources. The growing demand for water and over usage has depleted the resources, hence good governance and good management is necessary. A shift in perception towards ground water is needed more.

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